Sri Lanka Institute of Information Technology

Faculty of Computing, Department of Computer Systems Engineering

IE3122: Network Security

Week 10 Tutorial Worksheet 2: Deploying and Configuring AAA

Field	Detail
Based on	Week 10 Deck B, AAA Deployment lecture (Local vs Server-Based AAA, Device Administration vs Network Access AAA, RADIUS vs TACACS+, method lists, Cisco AAA configuration)
Learning Outcomes	LO3 (analyse security requirements), LO4 (evaluate security technologies), LO5 (configure security technologies)
Format	13 structured questions, no MCQs. Answer in your own words. Several questions ask you to write or interpret real AAA configuration.
How to use this	Answer every question in the space provided. Together, your answers form a set of study notes for this topic. Use them directly when revising for the mid-semester test and final exam.

The final section's questions ask you to explain real Cisco configuration steps, which is deliberate: LO5 is specifically about configuring a security technology to satisfy a requirement, not just describing what it can do.

Local vs. Server-Based AAA

Q1. A small branch office with two routers is deciding between local and server-based AAA. Recommend which approach is more appropriate given this scale, and justify your answer using both cost and administrative overhead. *[LO3, LO4]*

Q2. Explain the specific new risk that server-based AAA introduces that local AAA does not have, and explain what this risk means in practice if it is left unaddressed. *[LO4]*

Device Administration vs. Network Access AAA

Q3. Explain the difference between device administration AAA and network access AAA in terms of what is actually being protected in each case. *[LO3]*

Q4. Explain why TACACS+'s design, full encryption and clean AAA separation, suits device administration specifically, while RADIUS's lighter, bundled design suits network access specifically. Use the actual stakes and scale of each scenario in your answer. *[LO4]*

RADIUS vs. TACACS+

Q5. Compare RADIUS and TACACS+ specifically on what portion of the packet each encrypts, and explain the practical security consequence of that difference. *[LO4]*

Q6. A university needs to authenticate thousands of students and staff connecting to its WiFi network simultaneously throughout the day. Recommend RADIUS or TACACS+ for this scenario, and justify your recommendation using at least two specific characteristics of the protocol you chose. *[LO3, LO4]*

Q7. An administrator argues that TACACS+ should be used for everything, including WiFi authentication, since it is more secure than RADIUS. Evaluate this argument, and explain what would actually go wrong if TACACS+ were deployed for a large-scale WiFi network. *[LO2, LO4]*

Q8. Explain why RADIUS being an open, multi-vendor standard matters practically in a network that includes equipment from multiple manufacturers, not just Cisco. *[LO4]*

AAA Method Lists

Q9. Explain what an AAA method list actually does, and explain the specific problem it solves that would otherwise exist in a purely server-based AAA deployment. *[LO3, LO4]*

Q10. Explain the connection between how an AAA method list is processed and how an ACL from Week 7 is processed. What do they have in common? *[LO4]*

Cisco Configuration

Q11. List the correct order of commands needed to configure TACACS+ server-based authentication with a local fallback on a Cisco device, and briefly explain what each command actually does. *[LO1, LO5]*

Q12. In the command aaa authentication login default group tacacs+ local, identify which part of this command defines the method list's fallback order, and explain what would happen if the order were written the other way around. *[LO5]*

Q13. Explain what the aaa new-model command actually does, and explain why every other AAA configuration command in this week's material depends on it being entered first. *[LO5]*

Turning This Into Exam Revision

Once you've answered every question, you have the beginnings of a proper study-notes set for this topic. To make it exam-ready:

1. Reread each answer and check it would make sense to you in three months, without the lecture slides open.
2. The final section's questions are close to what you'll actually type in a lab. If you can answer them without looking at the slides, the hands-on work will go much faster.
3. Keep this worksheet with your notes for the mid-semester test (MCQ, LO1 and LO2) and final exam (MCQ + structured essay, LO1 through LO5).